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B.E - (Computer) (Sem.-VII)(Revised Course 2019-2020)

EXAMINATION JUNE 2023

Data Analytics

[Time: 03:00 Hours]

[Max. Marks:100]

Instructions:1. Assume suitable data.

2. Answer any 02 questions from PART- A and PART- B each and Attempt 01 question from PART-C.

PART A**Q1 a)** Calculate χ^2 value using the table below. Critical $\chi^2 = 7.851$.**(7 marks)**

	Observed Frequency		Expected Frequency	
	Female	Male	Female	Male
Without Graduation	6	7	6.08	6.92
College	13	16	13.55	15.45
Bachelor's Degree	16	15	14.49	16.51
Master's Degree	8	11	8.88	10.12
Total	43	49	43	49

b) A survey of cars was conducted in 2011 and determined that 60% of car owners have only 1 car, 28% have two cars and 12% have 3 or more cars. Supposing that you have decided to conduct your own survey and have collected the data below, determine whether your data supports the results of the study. Use a significance level of 0.05(5%). Also given that out of 129 car owners, 73 have one car and 38 have 2 cars. Critical value = 5.99. **(7 marks)**

c) From the given data in the following table, find out whether there is any relationship between gender and performance of the color. Critical value ($\chi^2_{0.05}$) = 5.991. **(6 marks)**

Color	Male	Female
Red	25	45
Blue	45	25
Green	50	10
Total	120	80

- Q2 a)** In an anti-malaria campaign in a certain area quinine was administered to 812 people out of a total of 3248. The number of fever cases is shown below. $X^2_{0.05} = 3.84$. **(5 marks)**

Treatment	Fever(A)	No Fever(a)	Total
Quinine(B)	140	30	170
No Quinine(b)	60	20	80
Total	200	50	250

- b)** Write short notes on logistic regression, ANOVA, Hypothesis test and Probability of error. **(4 marks)**
- c)** Find the value of t-test value for the following set of values: 7, 2, 9, 8 and 1, 2, 3, 4 **(4 marks)**
- d)** Internal test marks of students are as follows: 10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9. Mean population score is 12 and standard deviation is 1.80. Is the mean value for student significantly differed from the mean population value? Let alpha value is 0.05, so corresponding t value is 2.262. **(7 marks)**
- Q3 a)** A sample size of 16 people is taken. Suppose a claim is made that the average number of days a person spends a vacation is more than or equal to 5 days based on sample of 16 people whose mean came out to be 9 days. Sample standard deviation is found to be 3 days at a level of significance of 0.05 and t value for right-tailed test comes out to be 1.753. Calculate the value of t and check null hypothesis. **(6 marks)**
- b)** The marks of boys and girls are as follows: **(6 marks)**
 Boys: 12, 14, 10, 8, 16, 5, 3, 9, 11
 Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 15
 Is there any significant difference between the marks of girls and boys? Assume $\alpha=0.05$, t-value (critical) = 2.11.
- c)** To assess the significance of possible variation in a certain test between the convent schools of a city, a common test was given to a number of students taken at random from the fifth class of 3 schools concerned, the results are given below. $F_{0.05}=3.89$. **(8 marks)**

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Make analysis of variance for the given data.

PART - B

- Q4 a)** Explain each phases of data analytic life cycle. **(7 marks)**
b) Compare MySql with NoSql database. **(6 marks)**
c) Explain the different data types supported in R with the help of examples. **(7 marks)**
- Q5 a)** Explain different types of NoSql database. **(5 marks)**
b) Explain different ways of reading data in R. **(5 marks)**
c) Differentiate between structured and unstructured data. **(5 marks)**
d) Explain how a decision tree model is built using R programming language. Also show commands to plot a tree. **(5 marks)**
- Q6 a)** Explain the following function with the help of examples: **(10 marks)**
(i) length()
(ii) count()
(iii) mean()
(iv) var()
(v) sd()
(vi) Median()
(vii) min()
(viii) max()
(ix) quantile()
(x) IQR()
- b)** Explain the function available in R programming language used to determine the relation between two attributes. **(5 marks)**
- c)** Given a list of values for Year and Sales Record attribute as shown below, create a Linear, regression model in R programming language and predict the sales record for the year 2035. **(5 marks)**
Year<- c(2005, 2009, 2013, 2017, 2021)
SalesRecord<-c(35, 50, 65, 75, 76)

PART – C

- Q7 a)** Explain database system structure with a neat diagram. **(7 marks)**
b) Explain at least three data visualization function in R programming language with the help of an example. **(6 marks)**
c) Explain different types of database languages. **(7 marks)**

- Q8 a)** The 3 samples below have been obtained from normal populations with equal variances. Test the hypothesis that the sample means are equal. $F_{0.05}=3.68$. **(8 marks)**

A	B	C
8	8	17
10	6	10
7	11	12
14	8	12
11	8	15
16	13	12

- b)** Discuss in detail Chi² test with detail example. **(2 marks)**
c) The IT1 marks of SE Computer Boys and Girls are as follows: **(5 marks)**
Boys: 12, 13, 14, 10, 18, 16, 5, 3, 9, 4, 11; Girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 4, 15.
d) Write short notes on: decision trees, SVM, Naïve Bayesian classifiers, text analysis, Apriori Algorithm. **(5 marks)**